Math 3 Name:_____

VII-4N Solve Equations Algebraically: Word Problems NOTES Date: _____ Per: ___

Solving Word Problems

Step 1: Create a verbal model

Step 2: Write an algebraic equation to match your model
- create a variable to represent any unknown quantities

Step 3: Solve

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1) You are participating in a walk for charity. Your aunt said she will give you $100. Your Uncle says he will give you $10 for every mile you walk. Your goal is to earn $270. Write and solve an equation to determine how many miles you need to walk.

Create a verbal model

Convert this model into an equation.

2) You have $18,500 in a savings account. You will save $500 each month. You want to save $20,000 so you can go on a VERY nice vacation. Write and solve an equation to determine how many months you need save for.

3) Bobsun and Bobzalez are each hired to build a fence around a rectangular park. Bobsun is told the longest side is equivalent to 10 yards less than ½ of the perimeter. Bobzalez is told that the longest side is 22 yards more than ⅖ of the perimeter. Write and solve an equation to determine the perimeter of the park.

4) In the 2024 NFC Divisional Round of the playoffs, the 49ers scored 15 more than ⅛ the total amount of points in the game . The Packers scored 6 more points than ⅓ the total amount of points in the game. Write an equation to determine the total points in the game.

5) Who won the game from problem 4?